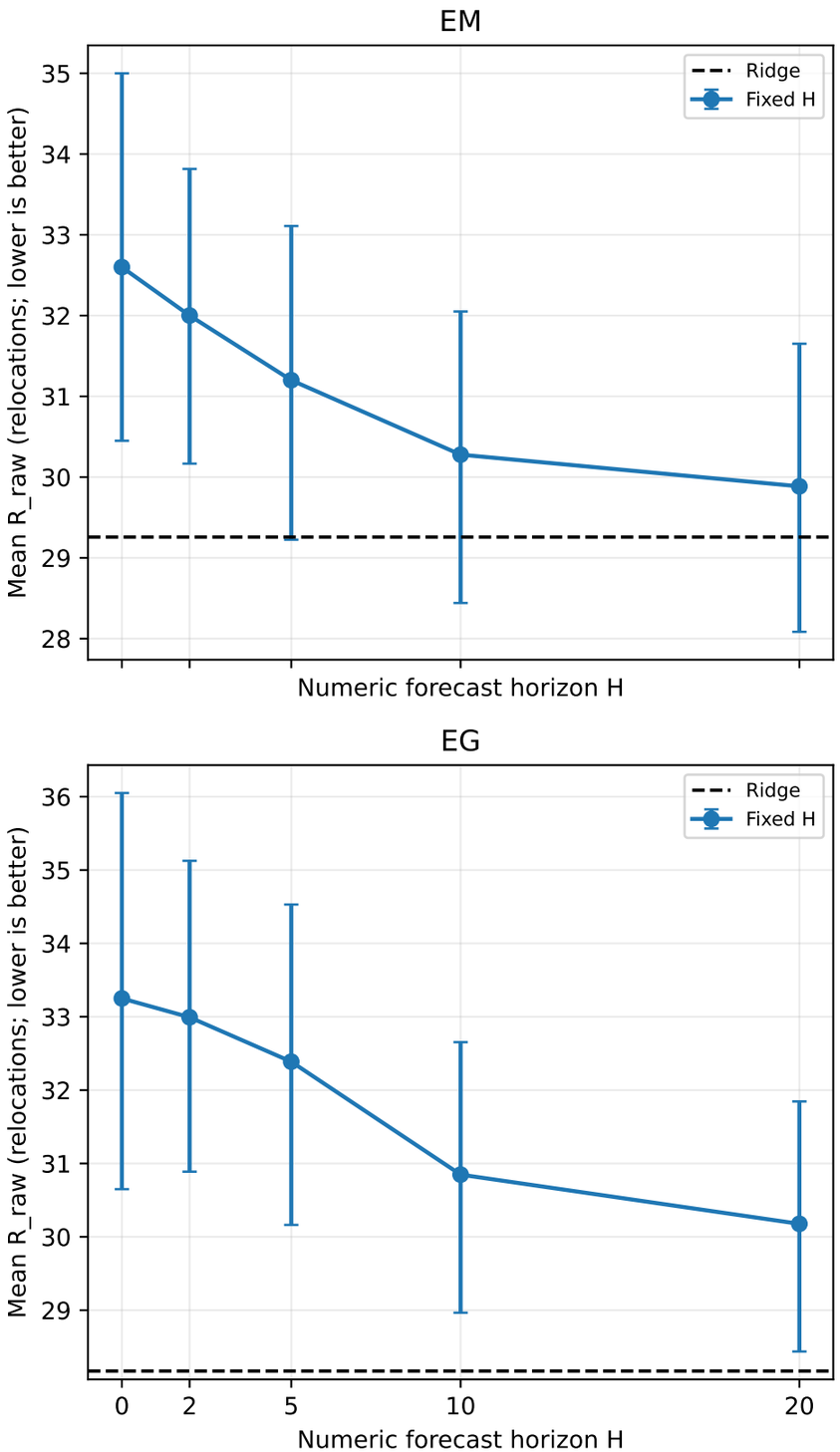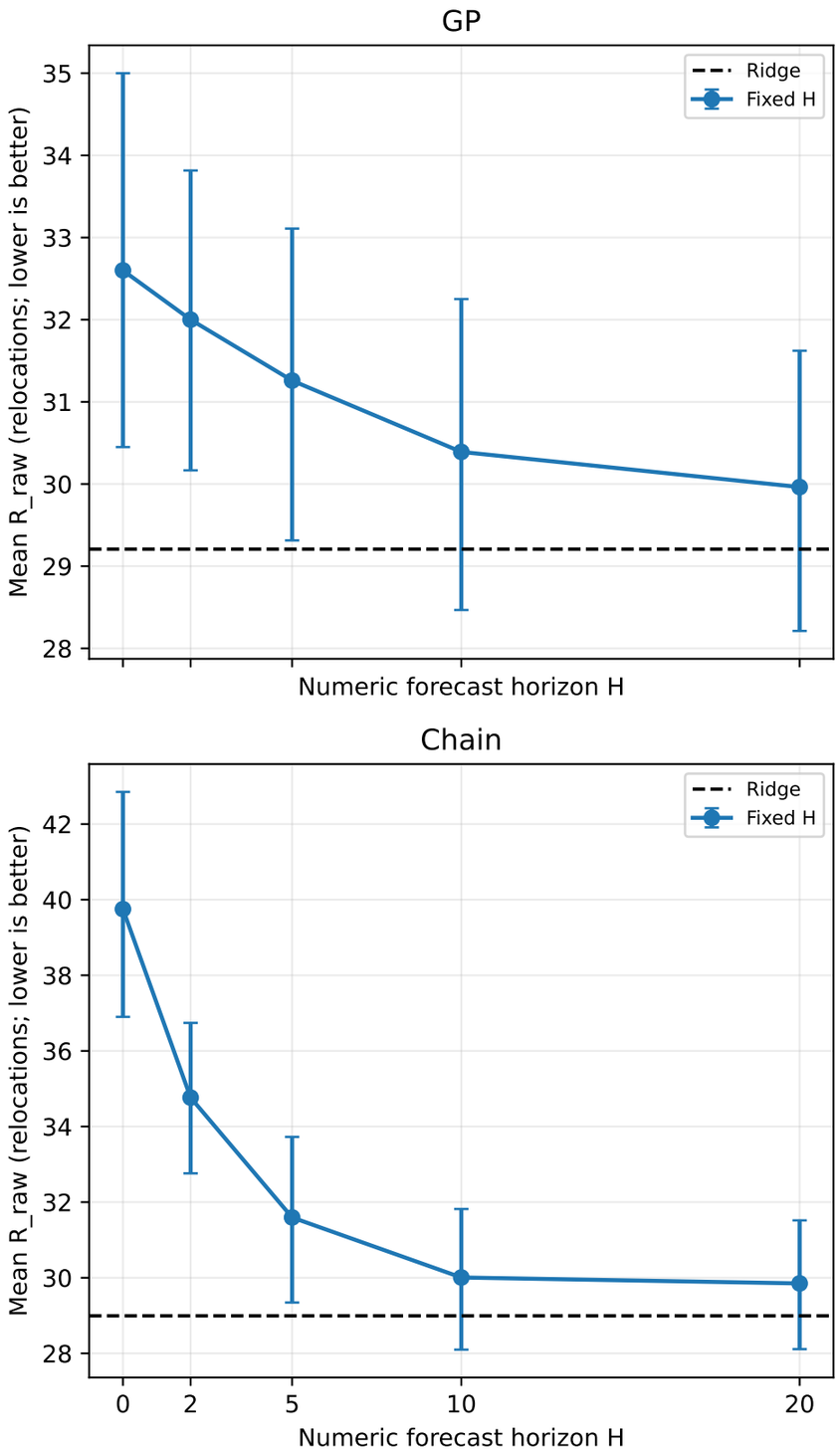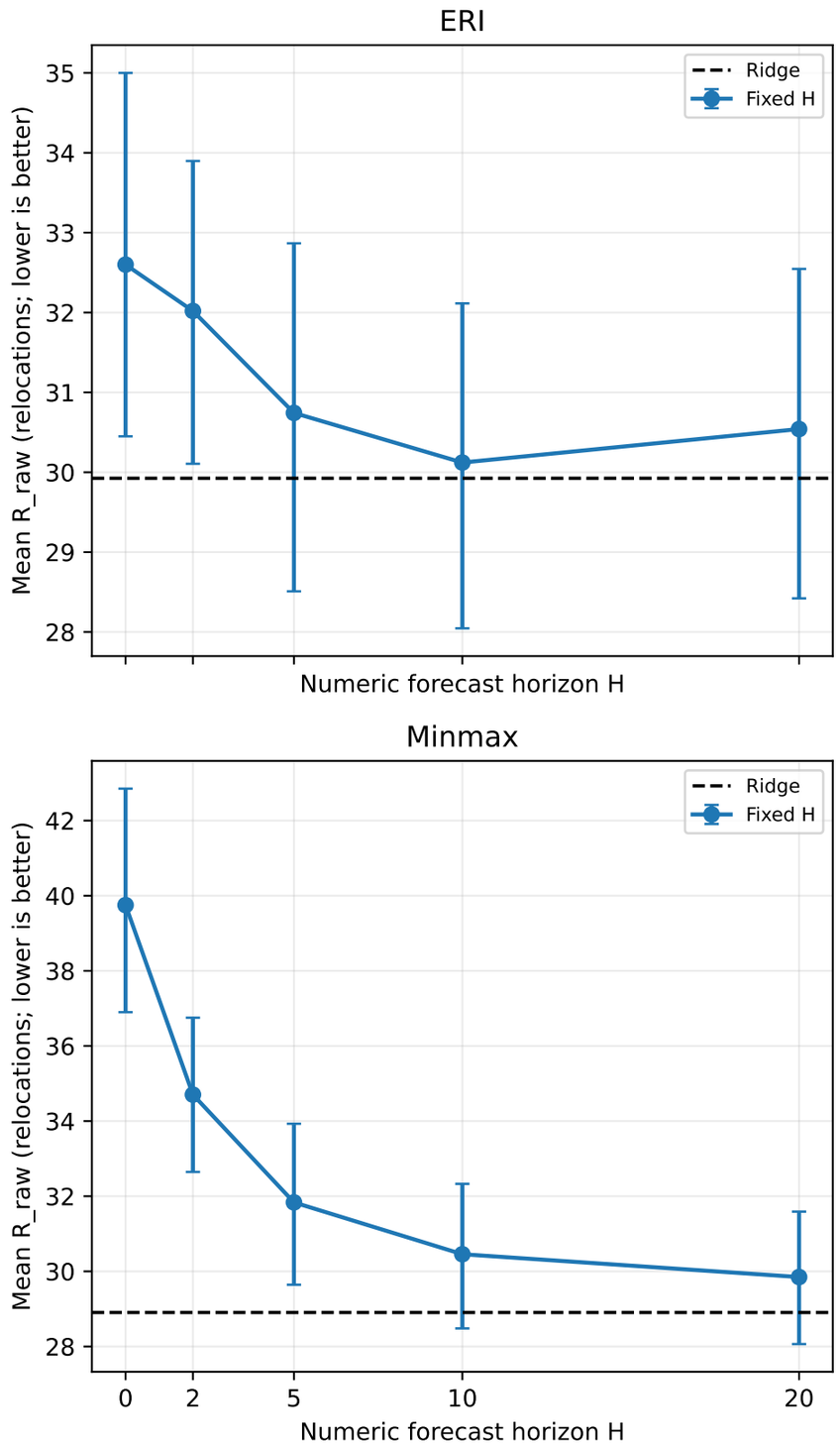Fixed-horizon relocation counts with Ridge reference; noisy macro